



Figure 11.2: *Arapaima gigas* displayed in the Siam Centre, Bangkok.

Arapaima is an important flagship genus for flooded forest ecosystem and human floodplain communities. Flagship taxa are used as a symbol to promote conservation awareness (Caro 2010). Their large size makes them a true freshwater megafauna like crocodiles, river dolphins, and other large fish. Freshwater megafauna face many threats, and 71% of these species are in decline (He et al. 2017, 2018). *Arapaima* continue to face intense fishing throughout their range (Watson et al. 2021). However, freshwater megafauna like the *Arapaima* have fewer conservation resources and efforts than marine or terrestrial megafaunas.

Fishing, in general, and fishing for *Arapaima* in particular, is a central element of the local economy and culture in Amazonia. Because these fish are **obligate** breathers, they are traditionally harvested by fishers using harpoons at the time when they surface to breathe. Men typically fish from canoes and search for signs of *Arapaima* near the surface. As they near the *Arapaima*, the harpooner throws the harpoon by hand. This is a specialized type of fishing, and the local fishers possess knowledge of the behavior that increases their likelihood of catching one. With appropriate training, fishers' participation in management processes can contribute to the conservation and governance of these small-scale fisheries.

Many populations of *Arapaima* have been driven to local extinction due to overfishing (Castello et al. 2015a; Gurdak 2019a; Watson et al. 2021; Freitas and Sousa 2021). Much of the catch is illegal, with most specimens being caught below the minimum size limit or during the closed season (Cavole et al. 2015). The small-scale fishers are geographically dispersed, and governments in these regions have insufficient resources to devote to enforcing fishing rules. The riverine fishers who target *Arapaima* are **marginalized** and have limited formal education. Yet, compliance with regulations is essential to prevent overfishing and local extinction.

Arapaima represent only a small fraction of the fisheries harvest, but they are culturally important and symbolic as a flagship genus of tropical South American fisheries and floodplain management and conservation. Reducing the threats to *Arapaima* will also provide protections for many of the highly migratory fish of the Amazon basin. Collectively, the migratory fish contribute most of the fishery's landings in the basin (Duponchelle et al. 2021). Migratory fish depend on multiple, distant, but interconnected habitats during their life cycle. Any threat to one of the habitats or the corridor that connects them can influence these important food fish (Goulding et al. 2019).